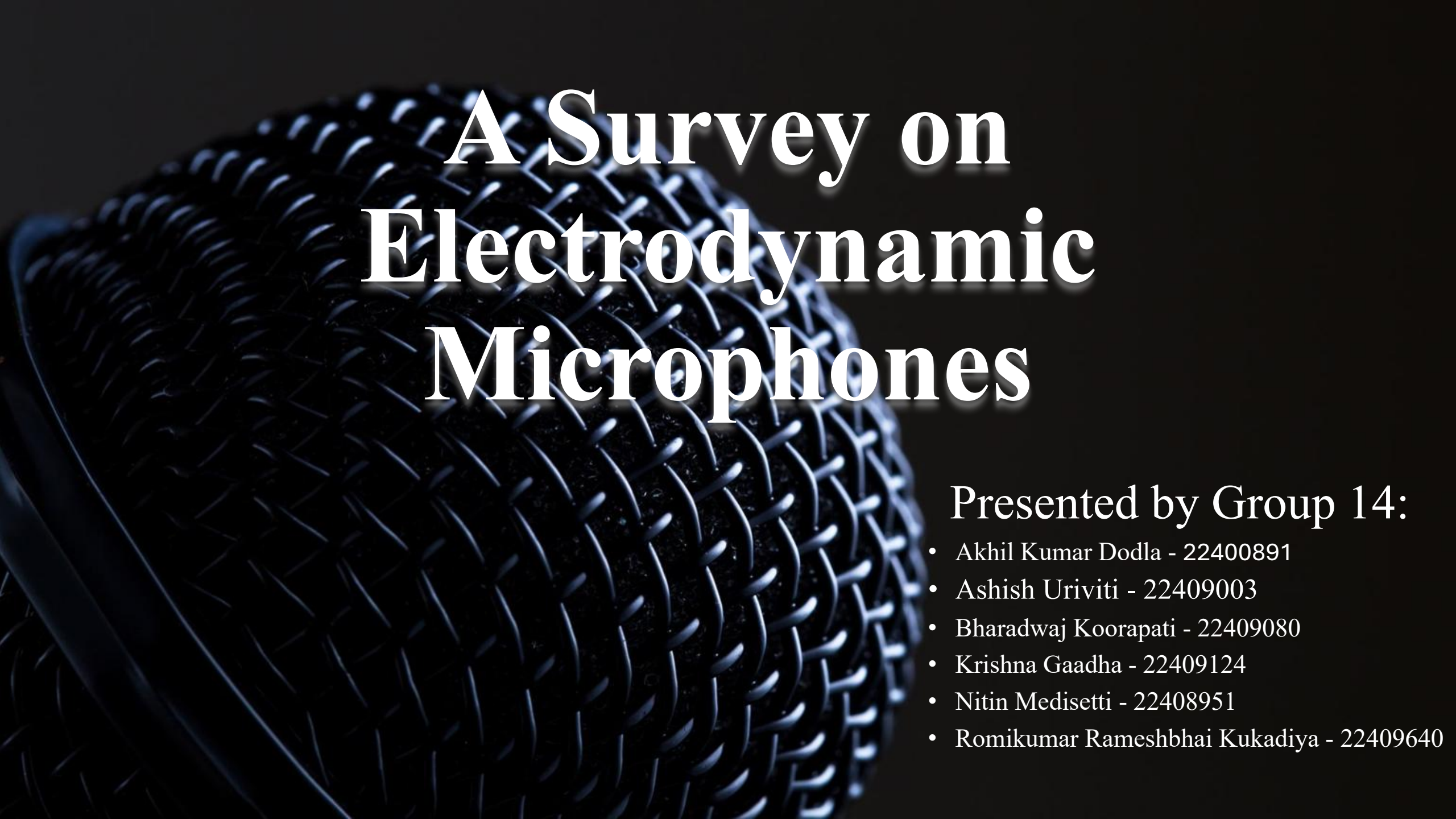




A Survey on Electrodynamic Microphones

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What are Electrodynamic Microphones

Electrodynamic microphones are a type of microphone that operates on the principle of electromagnetic induction to convert sound into an electrical signal.



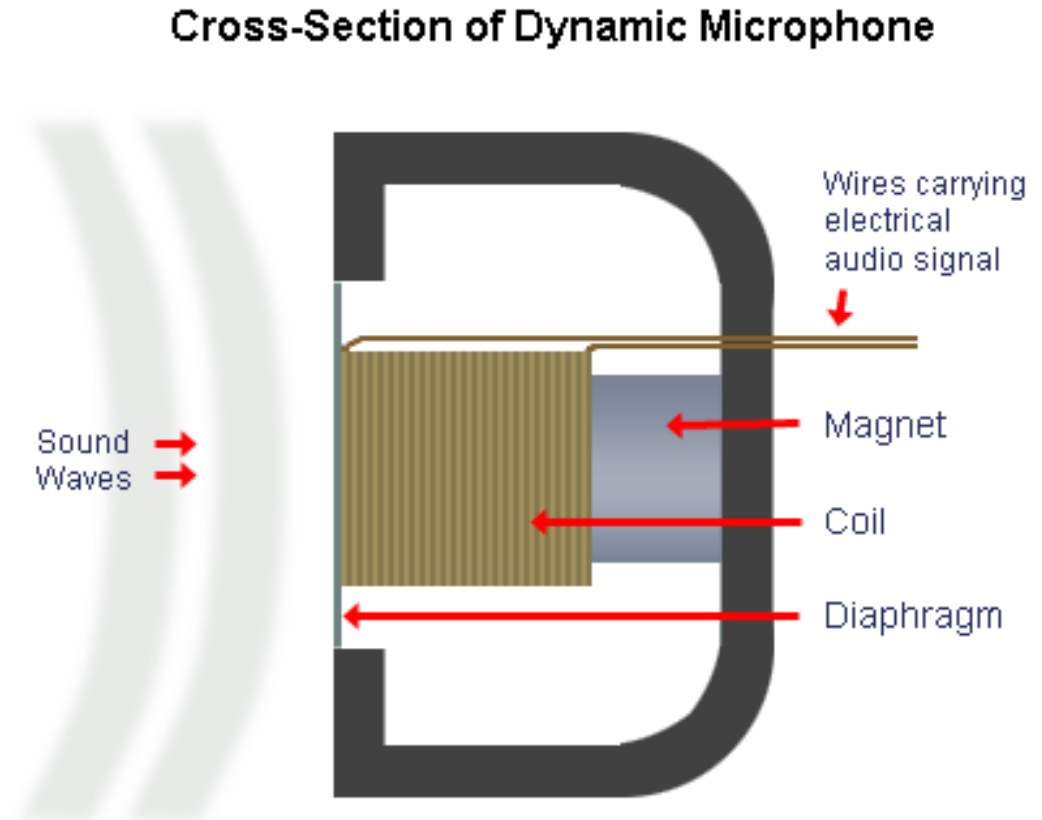
Motivation

- **Versatility:** Widely used in music and podcasting for balanced performance.
- **Durability:** Accurate sound capture with a simple, robust design.
- **Digital Era Role:** Key analogue input for preserving sound quality.
- **Future Scope:** Innovations ensure continued relevance and growth.



Key components

- **Diaphragm:** A thin flexible membrane that vibrates in response to sound waves
- **Coil:** A small lightweight coil of wire attached to the diaphragm, moves within a magnetic field.
- **Permanent magnet:** A magnet that creates a stable magnetic field around the coil
- **Enclosure (Housing):** A casing that protects the internal components and provides structural support
- **Output Terminals:** Carry the induced electrical signal to an external device

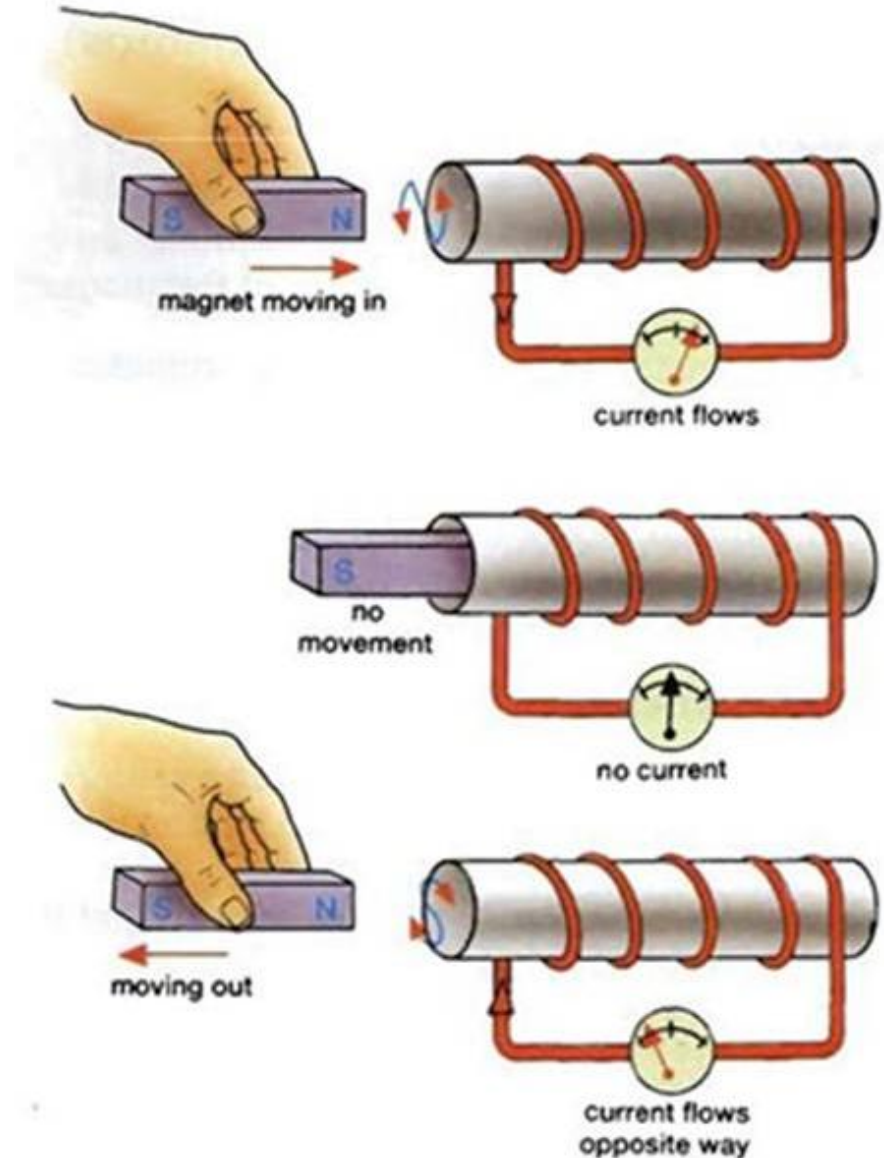


Source: [Connecting Dynamic Microphone Element - Forum - Circuits Online](#)

Working Principle

Faraday's Law of Electromagnetic Induction:

- This principle states that when a conductor moves within a magnetic field, an electric current is induced in the conductor



The Process of Sound-to-Electric Signal Conversion

Sound Wave capturing

- Sound waves hit the diaphragm, causing it to vibrate based on the sound's intensity and frequency

Diaphragm vibration

- The attached coil moves within the magnetic field of the permanent magnet, following the diaphragm's motion

Induction of Electric current

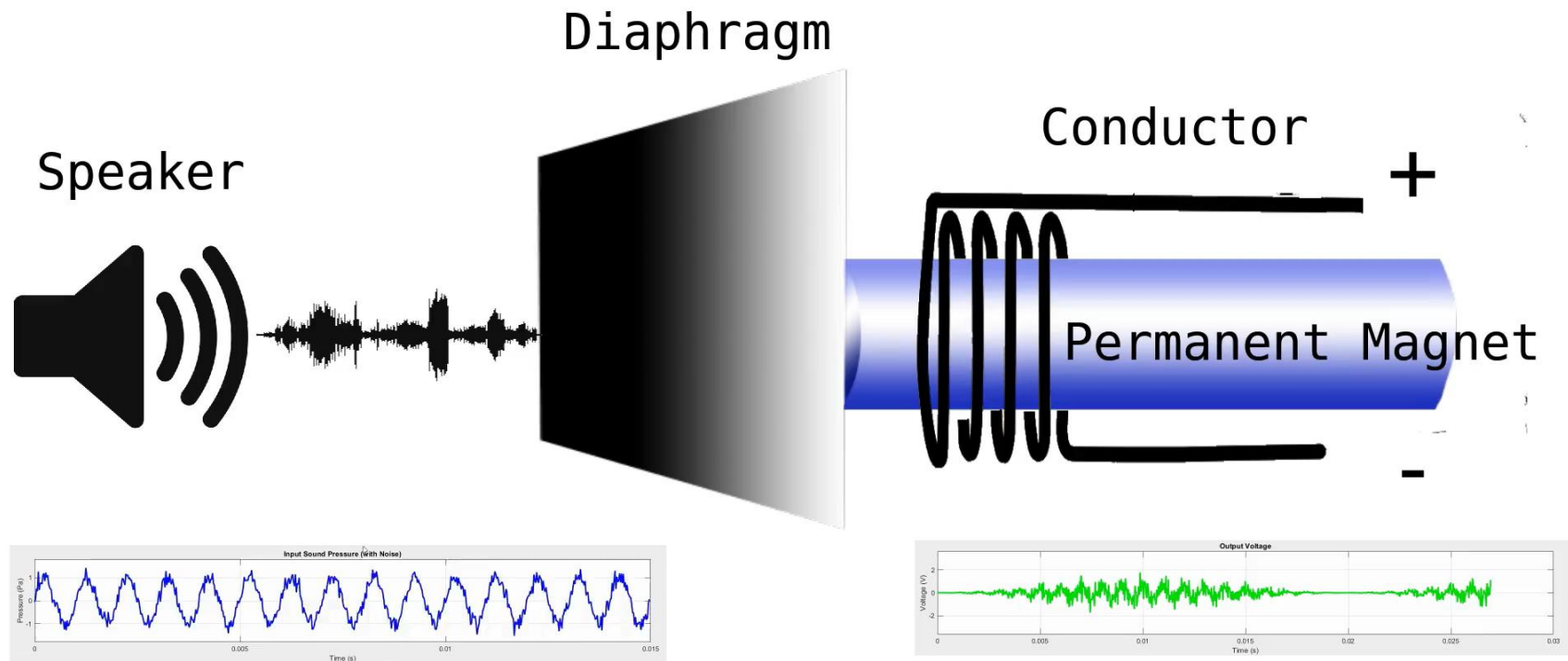
- The coil's movement through the magnetic field generates an AC signal as per Faraday's law of Induction

Electrical signal output

- The induced signal is sent to the output terminals for amplification and further processing

Animation

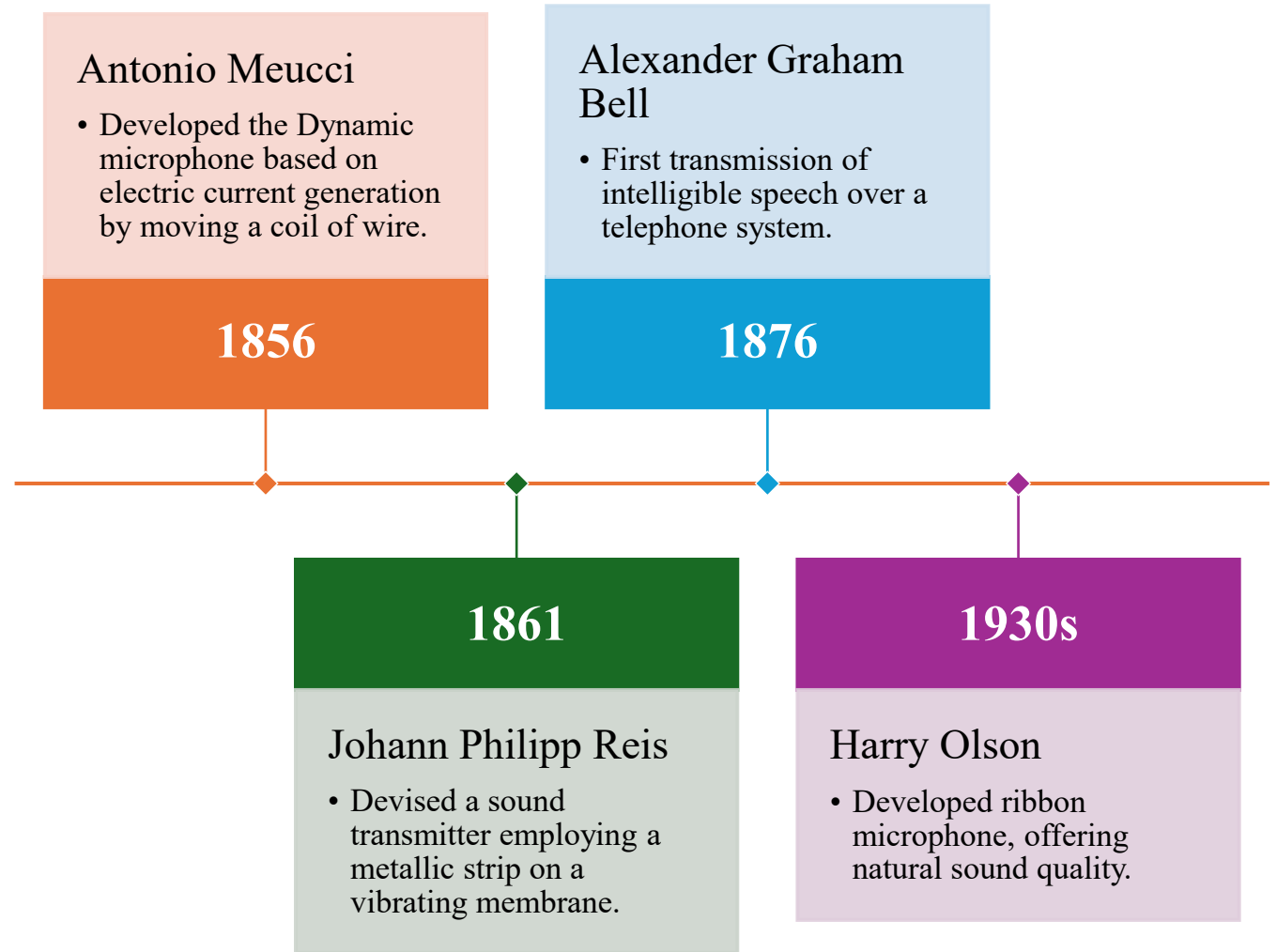
Visualisation of Electrodynamic Microphone



Literature Review



History of Microphones



Recent Advancements in Electrodynamic Microphone Technology

Use of Neodymium Magnets

- Allows compact and lightweight microphones without sacrificing sensitivity

Graphene membranes

- Ultra-lightweight, highly sensitive, broader frequency response and lower distortion.

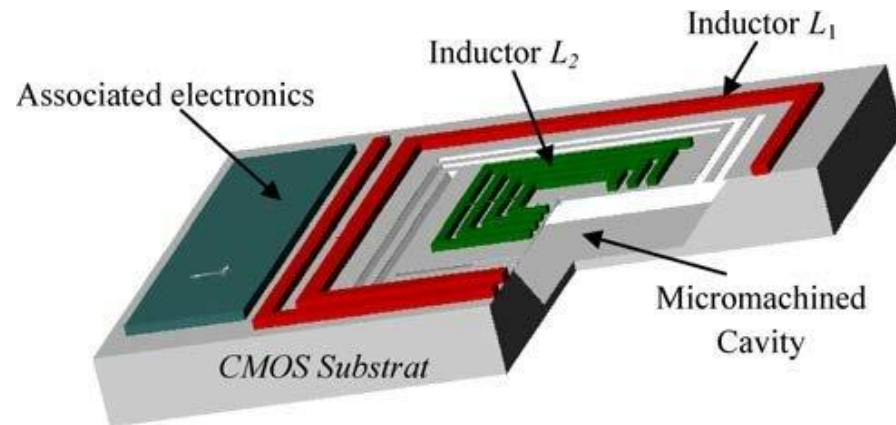
Optical Sensing Materials

- Achieve precise vibration detection, integrated into high-performance microphones.

Bio-inspired Materials

- Materials like artificial spider silk, provide extraordinary sensitivity and flexibility.

MEMS-Based Electrodynamic Microphones



A hybrid of traditional electrodynamic microphones and MEMS technology.



Combine the principle of electromagnetic induction with microfabricating.



Significantly smaller than traditional electrodynamic microphones.



Retains the durability and low noise of traditional dynamic microphones.



MEMS fabrication processes are intricate, requiring precise alignment of components.



Embedding strong magnets in MEMS designs without compromising performance is a technical challenge.



Used in smartphones, laptops, and wearable devices for high-quality sound capture.

Comparative Analysis



Electrodynamic microphone vs Condenser microphone vs Ribbon microphone

Features	Electrodynamic	Condenser	Ribbon
Durability	Exceptionally robust	Delicate due sensitive components	Can be damaged by air bursts or mishandling
Costs	Most affordable due to simple construction	Higher cost due to complexity	More expensive with premium features
Frequency response	Limited high-frequency response	Broad frequency range	Warm tone
Sensitivity	Less sensitive, ideal for louder sources like drums and amplifiers	Highly sensitive, can capture subtle details but prone to ambient noise.	Moderate sensitivity with smooth natural tones
Power source	No external power needed	Requires external power	May need external power in modern designs
Noise handling	Excellent isolation with directional polar patterns	More prone to picking up ambient noise	Similar to condenser but with smoother response

Applications of Electrodynamic Microphones



Live Sound Reinforcement:

Ideal for live performances, concerts, and events due to high sound pressure levels handling



Recording Studios:

Popular in studio settings for outstanding sound capture capabilities



Broadcasting

Widely used in television and radio studios for voiceovers, interviews, and presentations



Telecommunications:

Balancing sound quality and durability, suitable for portable communication devices.

Conclusion

Electrodynamic microphones remain an essential component of audio technology due to their versatility, and adaptability. As microphone technology advances, electrodynamic microphones are likely to retain their place as a foundational tool, balancing performance with practicality in diverse audio settings.



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Thank You

